

INTRODUCTION TO INFO. & COMM. TECHNOLOGY

18-10-2023

The Program Development Cycle

- The four steps to this plan are as follows:
 1. Completely understand the problem
 2. Devise a plan to solve it
 3. Carry out the plan
 4. Review the results

Program development

- Analyze the problem
- Design the program
- Code the program
- Test the program

Flowchart : a design tool

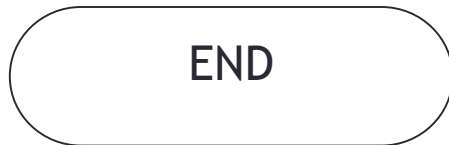
- A flowchart is a diagram that shows the “flow of control” of a program

Flowchart Symbols

- Rounded rectangles
- Parallelograms
- A rectangle
- Arrows(to show direction of flow)

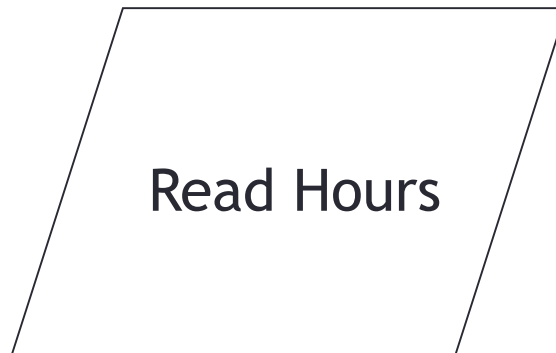
Symbols Contd.

- Terminals
 - represented by rounded rectangles/ ovals
 - indicate a starting or ending point of a program



Symbols Contd.

- Input/Output Operations
 - represented by parallelograms
 - indicate an input or output operation



Symbols Contd.

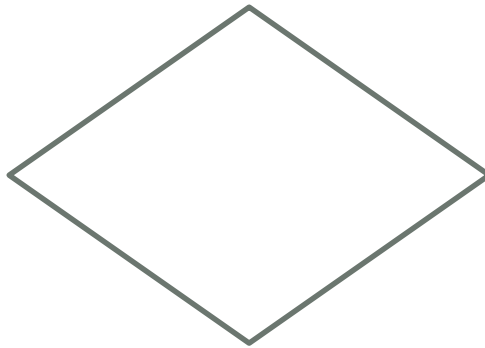
- Processes
 - represented by rectangles
 - indicates a process such as a mathematical computation or variable assignment

```
Sum = firstnum+ secondnum
```

```
LET Pi = 3.14
```


Symbols ..

- Decision
- Represents a program branch point

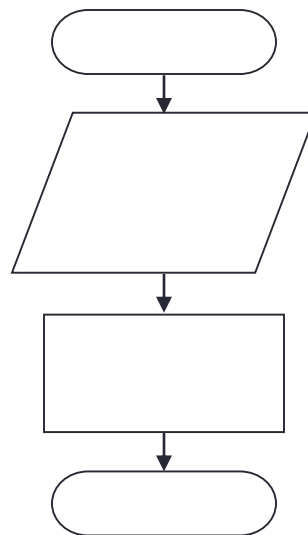


Flowchart Structures/Constructs

- Sequence
- Selection
- Iteration

Sequence Structure

- A series of actions are performed in sequence
- The sum-calculating flowchart of two given numbers → a sequence flowchart

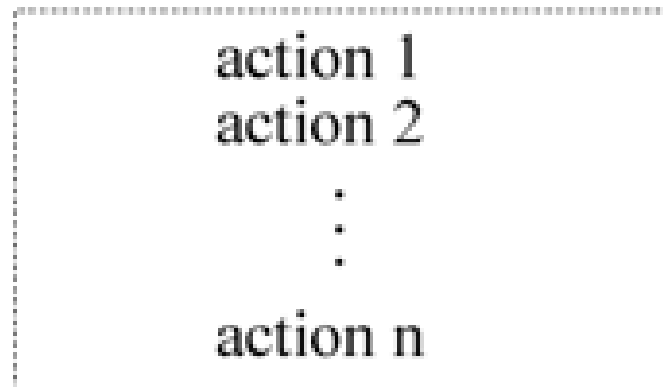


Example

- Write a program logic that accepts as input the mass in grams, and volume in grams per cubic cm, and outputs the density of the object using the formula: $\text{density} = \text{mass} / \text{volume}$
- Display the value of Density

Sequence Construct

- Code that executes from beginning to end
- Top to Bottom approach



a. Sequence

Practice Question - 1

- Draw a flowchart that reads in a number that represents the number of kilometers traveled
- The flowchart will convert this number to miles
 $1 \text{ kilometer} = 0.621 \text{ miles}$
- Display the kilometers travelled as well as miles travelled

Question - 1 (Solution)

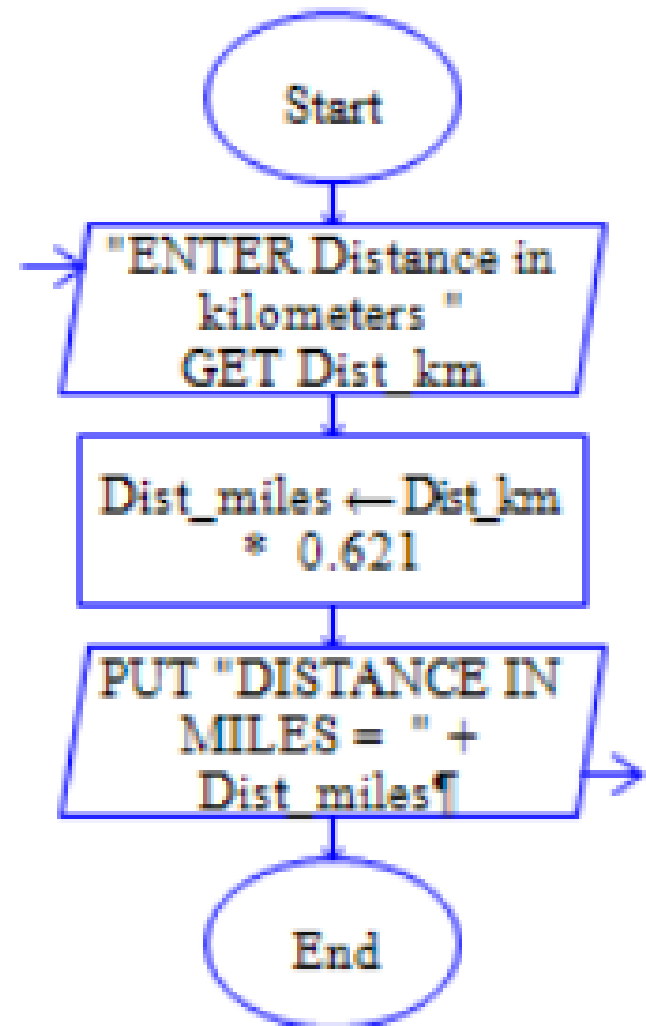
START

GET Dist_km

$\text{Dist_miles} = \text{Dist_km} * 0.621$

DISPLAY Dist_miles

STOP



Practice Question - 2

- Design a flowchart that will determine the area and perimeter of a rectangle
- Accept the length and width from the user
- Calculate and display the area and perimeter using the following formulae:

$$\text{Area} = \text{Length} * \text{Width}$$

$$\text{Perimeter} = 2(\text{Length} + \text{Width})$$

Question - 2 (Solution)

START

GET Length

GET Width

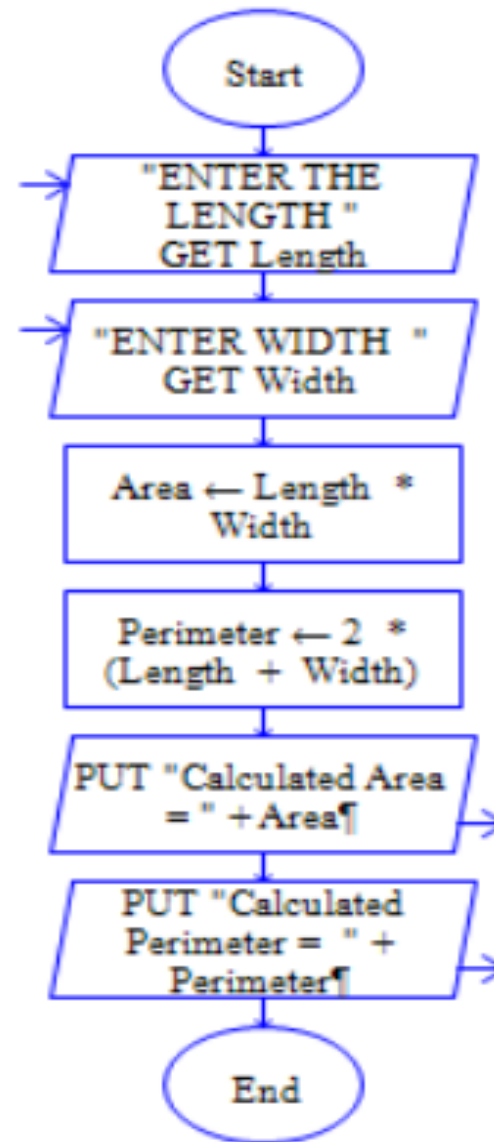
Area = Length * Width

Perimeter = 2 * (Length + Width)

DISPLAY Area

DISPLAY Perimeter

STOP



Practice Question - 3

- Design a flowchart that asks the user for his/her name and then displays a greeting message with the entered name as below:

Welcome to your First Coding Class SHAZIA

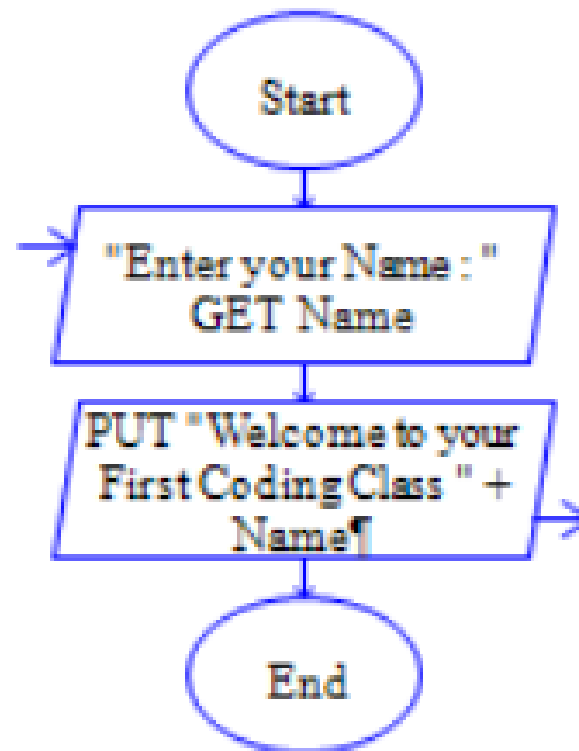
Question - 3 (Solution)

START

GET Name

DISPLAY "Welcome to your First Coding Class " + Name

STOP



Practice Question - 4

- Design a flowchart to print the sum, product, difference, and remainder after dividing the two numbers entered by the user.

Question - 4 (Solution)

START

GET num1

GET num2

sum = num1 + num2

product = num1 * num2

difference = num1 - num2

remainder = num1 Mod num2

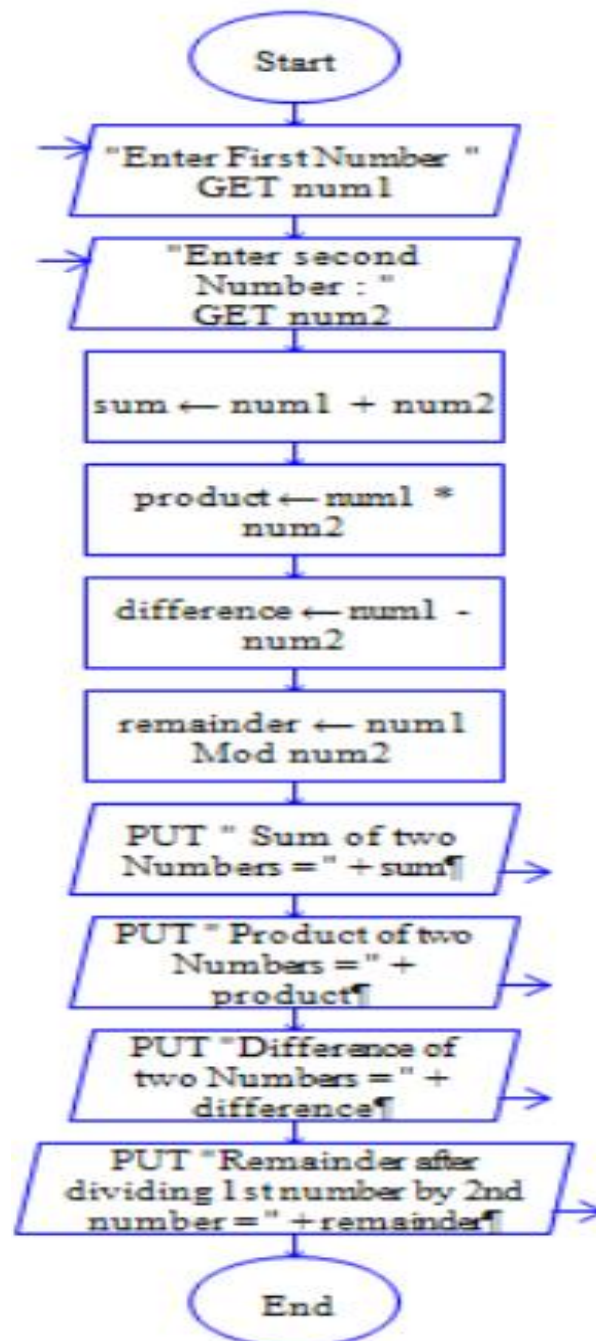
DISPLAY sum

DISPLAY product

DISPLAY difference

DISPLAY remainder

STOP



Practice Question - 5

- Design a flowchart to swap the two values entered by the user

For example the values entered by the user:

A = 10

B = 50

Values displayed by your code:

A= 50

B = 10

Question - 5

(Solution)

START

GET a

GET b

DISPLAY a

DISPLAY b

c = a

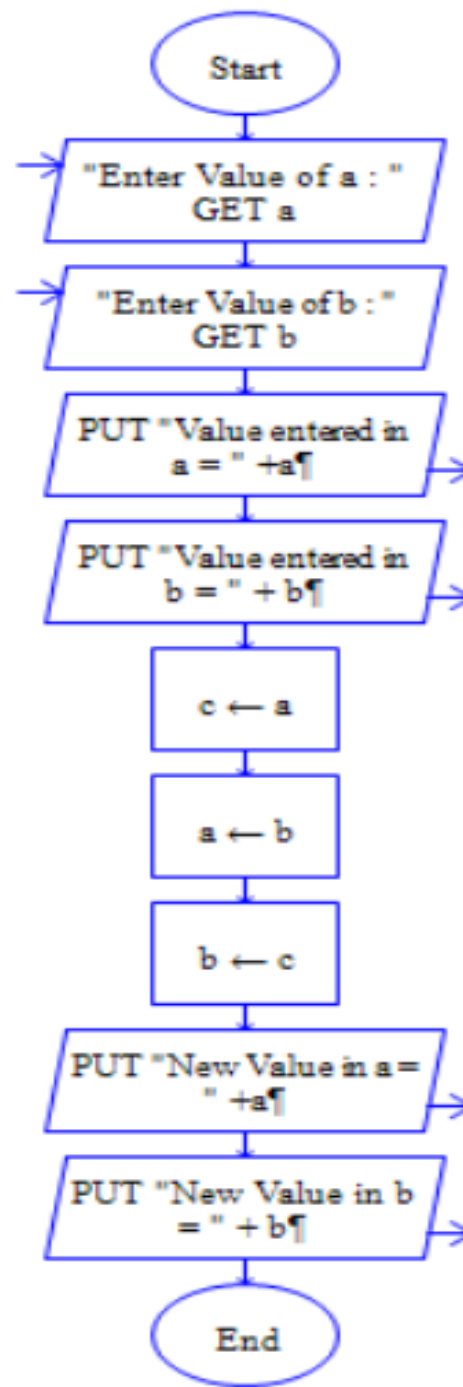
a = b

b = c

DISPLAY a

DISPLAY b

STOP



Practice Question - 6

- Design a flowchart to input temperature in degree Fahrenheit and to display after converting it to degree Centigrade.

$$^{\circ}\text{C} = (^{\circ}\text{F} - 32) * \frac{5}{9}$$

Question - 6 (Solution)

START

GET T_F

$T_C = (T_F - 32) * (5/9)$

DISPLAY T_F

DISPLAY T_C

STOP

Practice Question - 7

- Design a flowchart for a program that accepts number of remaining weeks till Quaid's Day.
- Your flowchart calculates the number of days left and displays weeks and days.

Question - 7 (Solution)

START

GET No_Of_weeks

No_Of_Days = No_Of_weeks * 7

DISPLAY No_Of_weeks

DISPLAY No_Of_Days

STOP

Practice Question - 8

- K&N sales division of Lahore generates 58 percent of total sales
- Based on that percentage, design a flowchart that will predict how much K&N sales division of Lahore will generate if the company has total of Rs. 8500,000 in sales this year.
- Display the Sales of Lahore division along with total sales of K&N.

Question - 8 (Solution)

START

Tot_Sales = 8500000

Sales_Lhr = Tot_Sales * (58/100)

DISPLAY Tot_Sales

DISPLAY Sales_Lhr

STOP

Practice Question - 9

- Design a flowchart that computes the total bill at the cash counter of a store.
- Assume the sales tax is 16.5 percent. Ask the user to enter the sales amount and your program logic should display the sales amount, sales tax, and total bill on the screen.

Question - 9 (Solution)

START

GET Sales_Amt

$\text{Tax} = \text{Sales_Amt} * (16.5/100)$

$\text{Tot_Bill} = \text{Sales_Amt} + \text{Tax}$

DISPLAY Sales_Amt

DISPLAY Tax

DISPLAY Tot_Bill

STOP

Practice Question - 10

- Design a flowchart that computes the tax and tip on a restaurant bill for a client
- Ask the user to enter meal cost.
- The tax should be 16.5 percent of the meal cost.
- The tip should be 15 percent of the total after adding the tax.
- Display the meal cost, tax amount, tip amount, and total bill on the screen.

Try to solve it yourself

Ref Chapters

- Book 1, Chp 11
- Book 2, Chp 8